

MARKET OBSERVATIONS—EMPIRICAL FINDINGS

In order that we better understand the meaning of these variability trends in the market, we evaluated volatility, correlation, and dispersion over the period January 1, 2000 through June 30, 2012. The universe used for our analysis was the SP500 index for large cap stocks and the R2000 index for small cap stocks. Stocks that were in each index on the last trading day in the month were included in calculations for that month.

Volatility: Volatility was computed as the 22-day standard deviation of log returns annualized using a scaling factor of $\sqrt{250}$. We computed volatility for all stocks in our universe on each day and reported the cross-sectional average as our daily data point. [Figure 6.2a](#) shows the average stock volatility for large cap stocks. At the beginning of 2000, volatility was fairly high. This was primarily due to the technology boom and increased trading volume. Volatility began the decade around 60% but decreased to about 20%. Volatility remained low until spring 2007 and then started to spike in August 2007 due to the quant crisis. But this was nothing compared to the financial crisis period of 2008–2009 where volatility spiked to over 100% and reached as high as 118%. Following the financial crisis volatility decreased to more reasonable levels of 20–30% and these lower levels persisted until fall 2011 due to the US debt ceiling crisis and the re-emergence of macroeconomic uncertainty in Europe. After the US addressed its debt issues volatility decreased shortly thereafter.

The average large cap volatility during our sample period was 36% and the average small cap volatility was 54%. Small cap volatility tends to be consistently higher than large cap volatility.

Correlation: Correlation was computed as the 22-day correlation measure across all pairs of stocks in each index. [Figure 6.2b](#) shows the average pair-wise correlation across all pairs of stock on each day for large cap stocks. For example, for a universe with 500 stocks there are 124,500 unique two pair combinations. Hence, we computed correlations across all 124,500 combinations of stock and show the average value in [Figure 6.2b](#). For an index consisting of 2000 stocks there are just slightly less than 2 million unique combinations of stock (1,999,000 to be exact). Computing average pair-wise correlations is a very data intensive process. The graph of average large cap correlations over our period found relatively low correlation (less than 30%) at the beginning of the decade but then spiking at the time of the collapse bubble. This increase in correlation was due in part to the markets re-evaluation of stock prices to lower levels and a sell-off of stock from institutions that further pushed the prices in the